UNITS 3–4 PRE-TEST

Equations, Inequalities & Angle Relationships

Standards assessed: NC.8.EE.7 · NC.8.G.5

Name: _____ **Date:** _____ **Period:** _____

SCORE **3.0** **Basic Understanding**

NC.8.EE.7 · NC.8.G.5 Solve one- and two-step equations and inequalities. Find angles with one step.

• I can solve simple equations and inequalities, and use angle pairs and the triangle sum to find a missing angle.

Match each vocabulary term to its definition. Then answer each question.

– ▾ transversal	**A.** Angles between the parallel lines, on the same side of the transversal. They add to 180°.
– ▾ corresponding angles	**B.** An equation that is never true, for any value of x.
– ▾ alternate interior angles	**C.** A line that crosses two or more other lines.
– ▾ same-side interior angles	**D.** Angles in the same position at each crossing. They are equal when the lines are parallel.
– ▾ exterior angle	**E.** An angle outside a triangle, made by extending one side.
– ▾ no solution	**F.** Angles between the parallel lines, on opposite sides of the transversal. They are equal.

Reference

Equal: corresponding, alternate interior, alternate exterior · Add to **180°**: same-side interior, linear pair, the 3 angles of a triangle

Exterior angle = sum of the two remote interior angles. Multiply or divide by a negative → flip the inequality sign.

1. *Solve*

$2x + 5 = 17$

x = _____

2. *Solve the inequality*

$x + 9 > 4$

Answer: e.g. x > 4

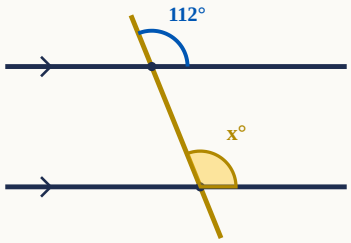
3. *How many solutions?*

$3x + 4 = 3x + 4$

Answer: e.g. no solutio

4. *Corresponding*

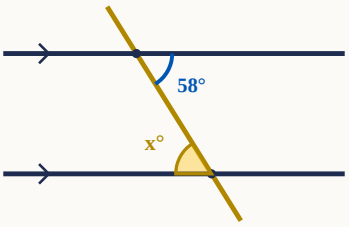
Parallel lines are cut by a transversal. Find the corresponding angle x.



x = _____

5. *Alt. interior*

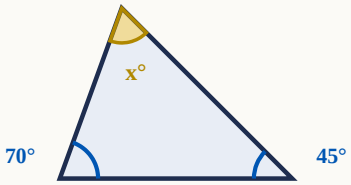
Parallel lines are cut by a transversal. Find the alternate interior angle x.



x = _____

6. *Triangle sum*

A triangle has angles 70° and 45°. Find the third angle x.



x = _____

SCORE **4.0** **Target Content**

NC.8.EE.7 · NC.8.G.5 Solve multi-step equations and inequalities. Use parallel lines and triangle theorems to find angles.

• *I can solve multi-step equations and inequalities, and use transversal angle pairs and the exterior angle theorem.*

Answer each question. Show your work.

7. *Solve*

$5x - 7 = 2x + 8$

x = _____

8. *Solve*

$3(x - 4) = x + 6$

x = _____

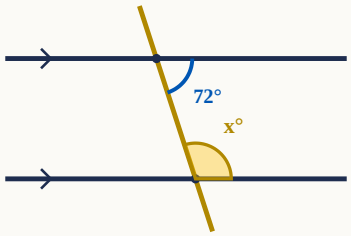
9. *Solve the inequality*

$-2x + 3 \leq 11$

Answer: e.g. x > 4

10. *Same-side interior*

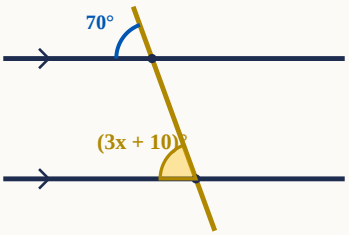
Same-side interior angles: one is 72°. Find the other angle x.



x = _____

11. *Solve*

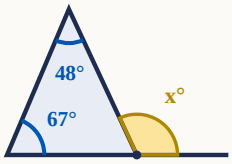
Corresponding angles measure (3x + 10)° and 70°. Solve for x.



12. *Exterior angle*

The remote interior angles are 48° and 67°. Find the exterior angle x.

x = _____



x = _____

SCORE 5.0 Advanced Content

NC.8.EE.7 · NC.8.G.5 Identify the number of solutions and solve multi-step angle problems with algebra.

• I can tell if an equation has one, no, or infinitely many solutions, and write and solve equations for angles.

Answer each question. Write "no solution" or "infinitely many" when that is the answer.

13. *How many solutions?*

$2(3x + 1) = 6x - 5$

Answer: e.g. no solutio

14. *Solve*

$4(x + 2) - x = 2(x + 7)$

x = _____

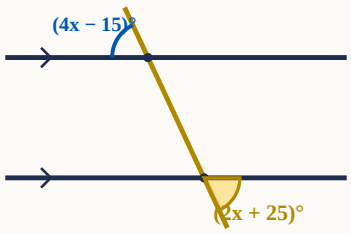
15. *Solve the inequality*

$3(x - 2) > 5x + 4$

Answer: e.g. x > 4

16. *Solve*

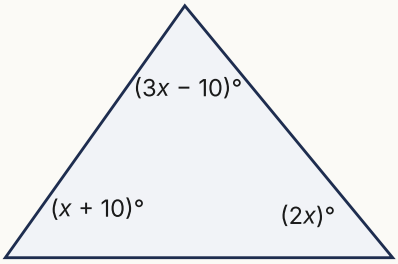
Alternate exterior angles measure $(4x - 15)°$ and $(2x + 25)°$. Solve for x.



x = _____

17. *Solve*

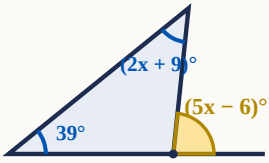
A triangle has angles (x + 10)°, (2x)°, and (3x − 10)°. Solve for x.



x = _____

18. *Solve*

An exterior angle measures (5x − 6)°. The remote interior angles are (2x + 9)° and 39°. Solve for x.



x = _____